Lecture 3

Processes & Scheduling

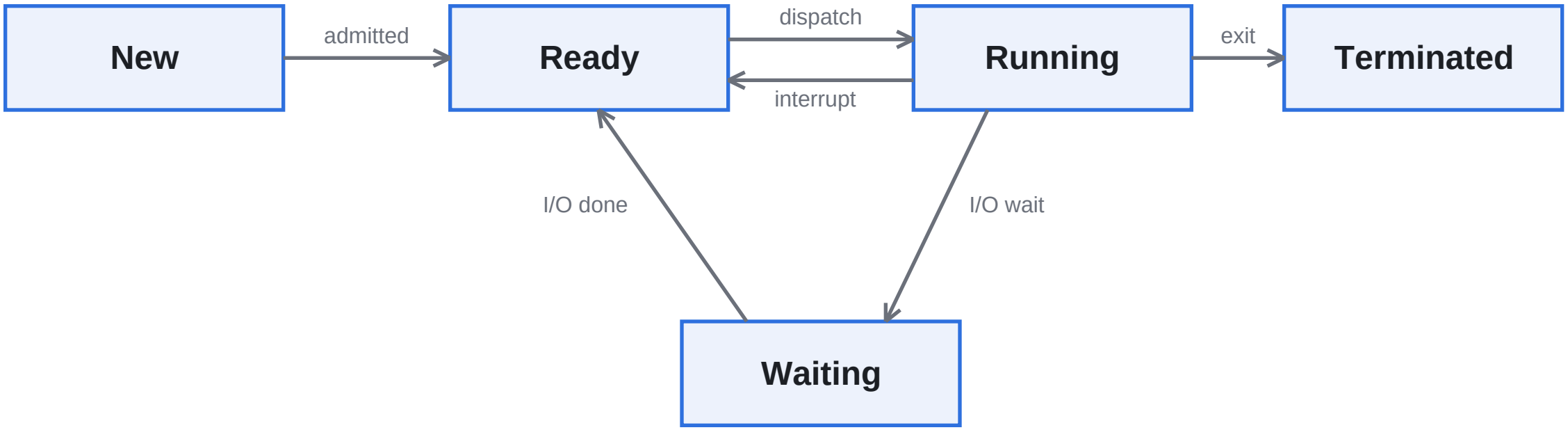
How an operating system decides what runs next

Prof. A. Example · Winter term 2026

What is a process?

- A process is a program in execution.
  - It has its own address space: code, data, heap and stack.
- The OS keeps a Process Control Block (PCB) for each one.
  - PCB: process ID, state, program counter, registers, open files.
- Many processes share one CPU by taking turns.
- Switching between them is called a context switch.

The five process states



Only one process can be Running per CPU core at any moment.

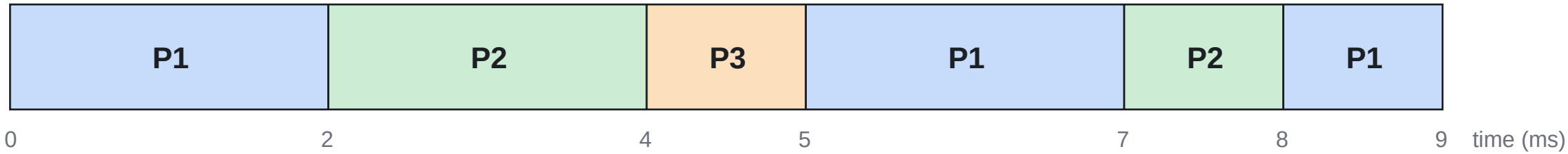
Processes vs. threads

	Process	Thread
Memory	Own address space	Shared with its process
Creation cost	High (fork, new PCB)	Low
Communication	IPC: pipes, sockets	Shared variables
Crash impact	Isolated	Can take down all threads

CPU scheduling algorithms

- FCFS: first come, first served. Simple, but long jobs block.
- SJF: shortest job first. Optimal average waiting time.
- Round Robin: each process gets a fixed time quantum.

Round Robin, quantum = 2



Summary

- A process = program + execution state (stored in the PCB).
- States: new, ready, running, waiting, terminated.
- Threads share memory, which makes them cheap but risky.
- The scheduler trades off fairness, throughput and response time.

Exam tip: be able to draw the state diagram and compute
average waiting time for FCFS, SJF and Round Robin.